

Group Number as in Blackboard: Group 4

Section Number: section 1

Group Members (5 students max):

1. Omar fawzi Hawsawi (2240007072) (Leader)
2. Abdulaziz khabled Alhassan (2240006848)
3. Nawaf Khaled Alwabel (2240001111)
4. Fahdel abbas al asaad(2240002427)
5. Mohammed Othman Alkhodir (2260040121)
6. Mohammed Nasser bin Ali Al-Marhoon (2240005893)

Definition of the problem:

The problem is taking the long path to arrive to you're destination so we will make a program to show you the shortest path possible the program will choose the path by making some calculations .

The algorithms that can solve this problem:

- Dijkstra's Algorithm
- Bellman-Ford Algorithm
- Floyd-Warshall Algorithm
- A* (A-Star) Algorithm

Among these algorithms the best one is Dijkstra algorithm, because it's the fastest, by the fastest I mean it complexity on the best case using **priority queue** is $\Omega(n \log n)$ and for the worst case using **simple array** is $O(n^2)$, while the other logarithms their best case complexity is $\Omega(n)$ and the worst case go to $O(n^2)$ and worse.

Pseudocode of the selected algorithms:

Algorithm ShortestPath(G, source)

1. node n in graph G
- 2.start a for loop
3. distance[n] = infinity (we don't know the distance)
4. visited[n] = false
5. End For loop

6. distance[source] $\leftarrow$ 0 (at the start of the map)

7. Entering a for loop to visit all the nodes
8. Choose the unvisited node u with the smallest distance
9. Mark u as visited
10. For each neighbor n of u
11. If distance[u] + weight(u,n) < distance[n]
12. distance[n] = distance[u] + weight(u,n)
13. End If
14. End For loop

15. End Repeat

16. Return distance